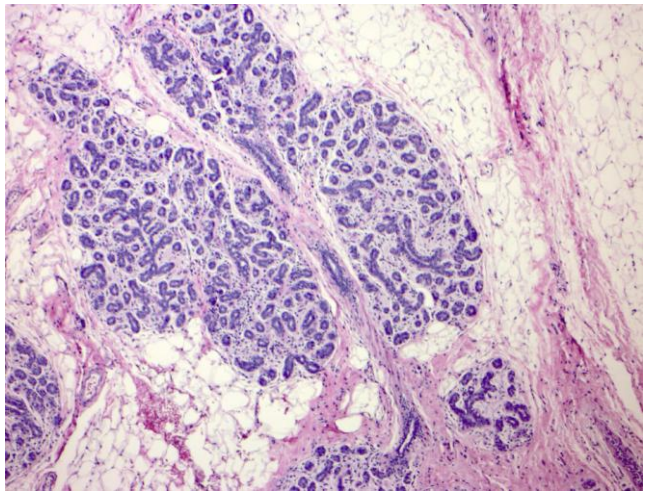
Spatial Gene Expression Prediction from Breast Histology

Group 5 | Project F | October 15, 2026

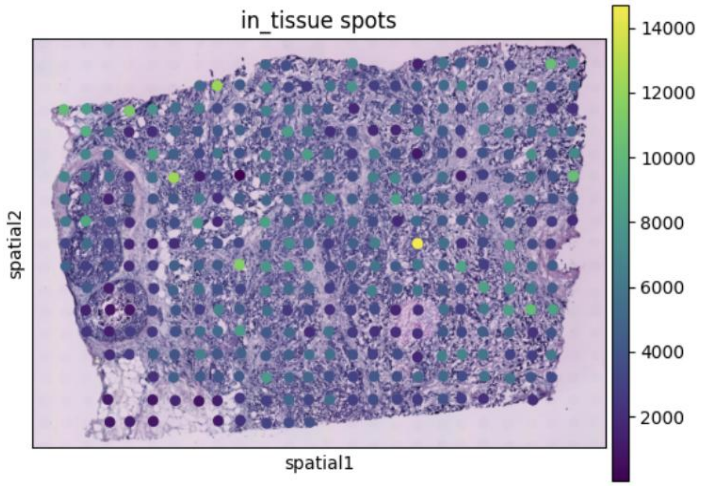
Mobai (James) Jiang, Junewoo (June) Koo, Siyu (Bobby) Wang, Jiayin (Moria) Li

JOHNS HOPKINS UNIVERSITY

Quick Recap



H&E Histology



Spatial Transcriptomics

Background

- H&E images capture tissue morphology, while spatial transcriptomics measures localized gene expression.
- Pretrained pathology models provide high-dimensional image embeddings for gene expression prediction.

Central Question for this Week

Does PCA reduce embedding dimensionality and computational cost while preserving gene expression prediction accuracy?

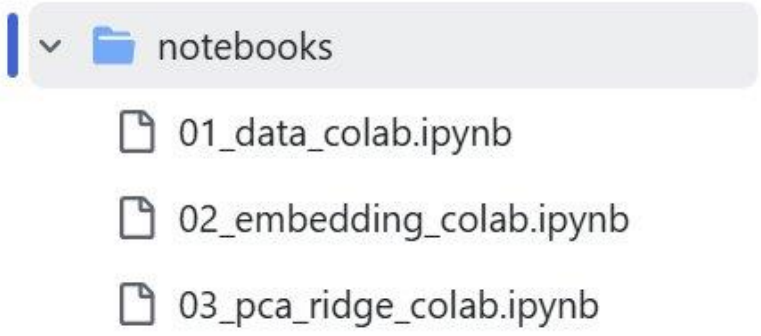
Overall Pipeline & Current Progress

HEST-1k Dataset (H&E patches + spatial gene expression)

01

Data Preparation

- Sample filtering
- Barcode matching
- Expression processing



02

Feature Extraction

- GenBio-PathFM
- Frozen image embeddings

03

PCA + Ridge Baseline

- Dimensionality reduction
- Gene prediction
- Evaluation framework

Current Status

- Three notebooks implemented and successfully tested.
- Preliminary PCA ablation completed on one slide.
- Full-cohort processing and comprehensive evaluation are pending.

Data Preparation & Alignment

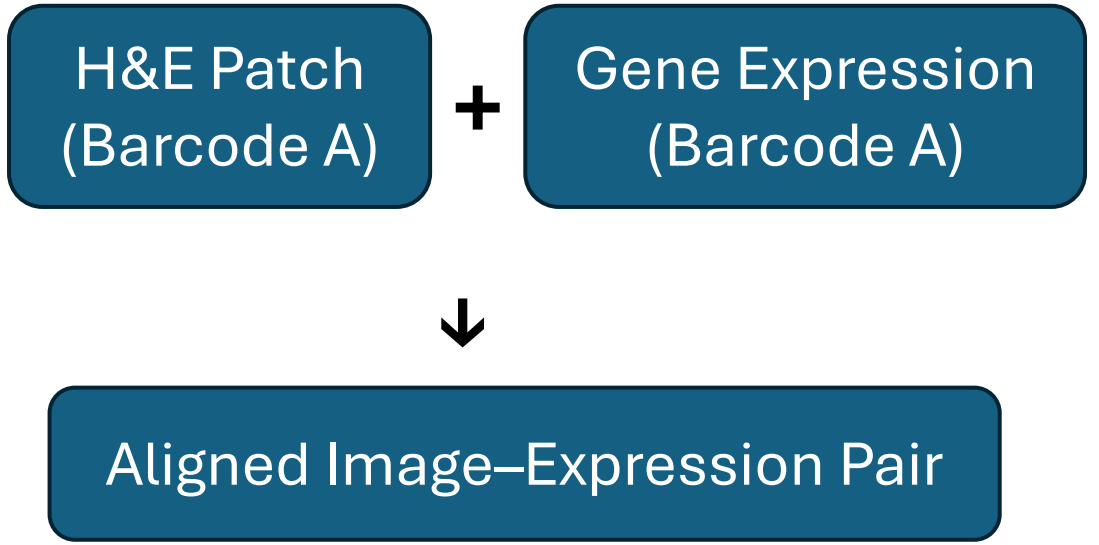
Notebook 01 — Establishing aligned image–expression pairs

1. Data Selection

- Load HEST-1k metadata.
- Filter human breast tissue from the selected ST study.
- Download H&E patches and expression matrices.

2. Barcode Alignment

- Match image patches to expression spots.
- Preserve consistent spot ordering.



Data Preparation & Alignment

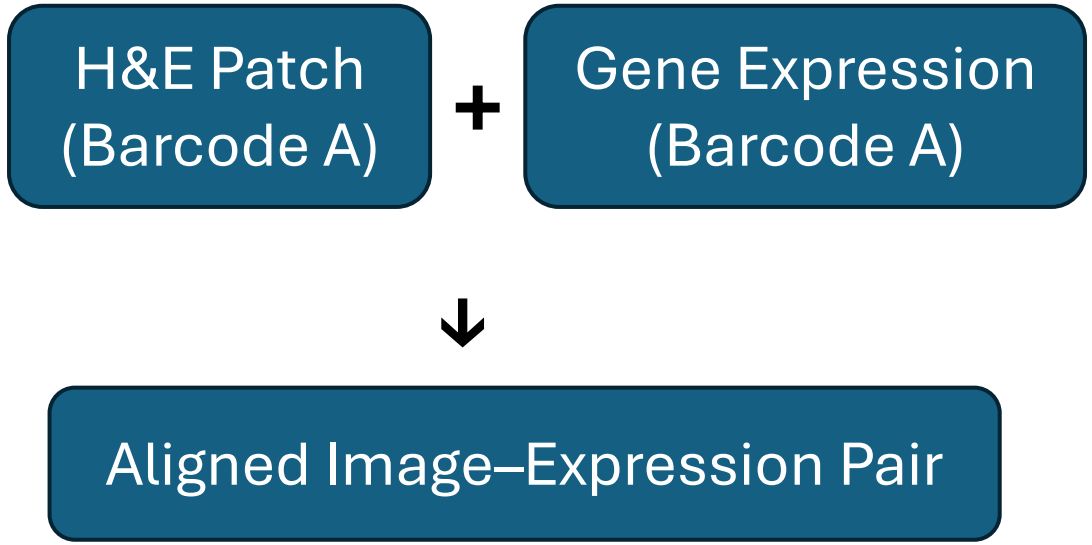
Notebook 01 — Establishing aligned image–expression pairs

3. Gene Processing

- Identify genes with high-variance.
- Apply log1p transformation.
- Save aligned expression matrices.

4. Quality Control

- Verify spot counts and barcodes.
- Visualize spatial spot distributions.

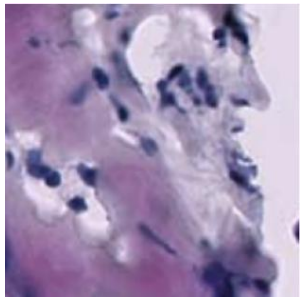


Feature Extraction with GenBio-PathFM

Notebook 02 — Converting histology patches into frozen image embeddings

Implementation

- Image normalization and batch inference.
- Batch size: 64.
- Save embeddings by section with corresponding barcodes.



H&E Patch (224 × 224)

↓

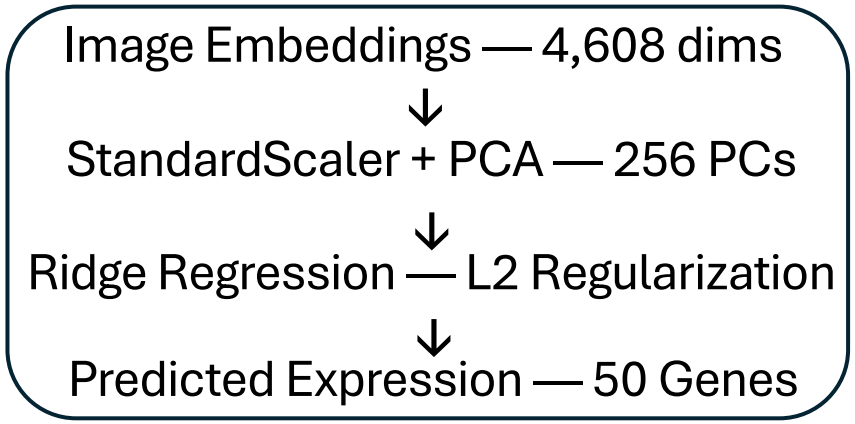
Pretrained GenBio-PathFM
Frozen model · GPU inference

↓

4,608-Dimensional Embedding
One feature vector per spot

PCA + Ridge Regression Baseline

Notebook 03 — Predicting spatial gene expression from frozen embeddings



Modeling

- Select top 50 variable genes using training data.
- Reduce embedding dimensionality.
- Tune Ridge regularization.

Evaluation Framework

- Spot-level split
- Slide-level split
- Patient-level split
- PCC, RMSE, NRMSE

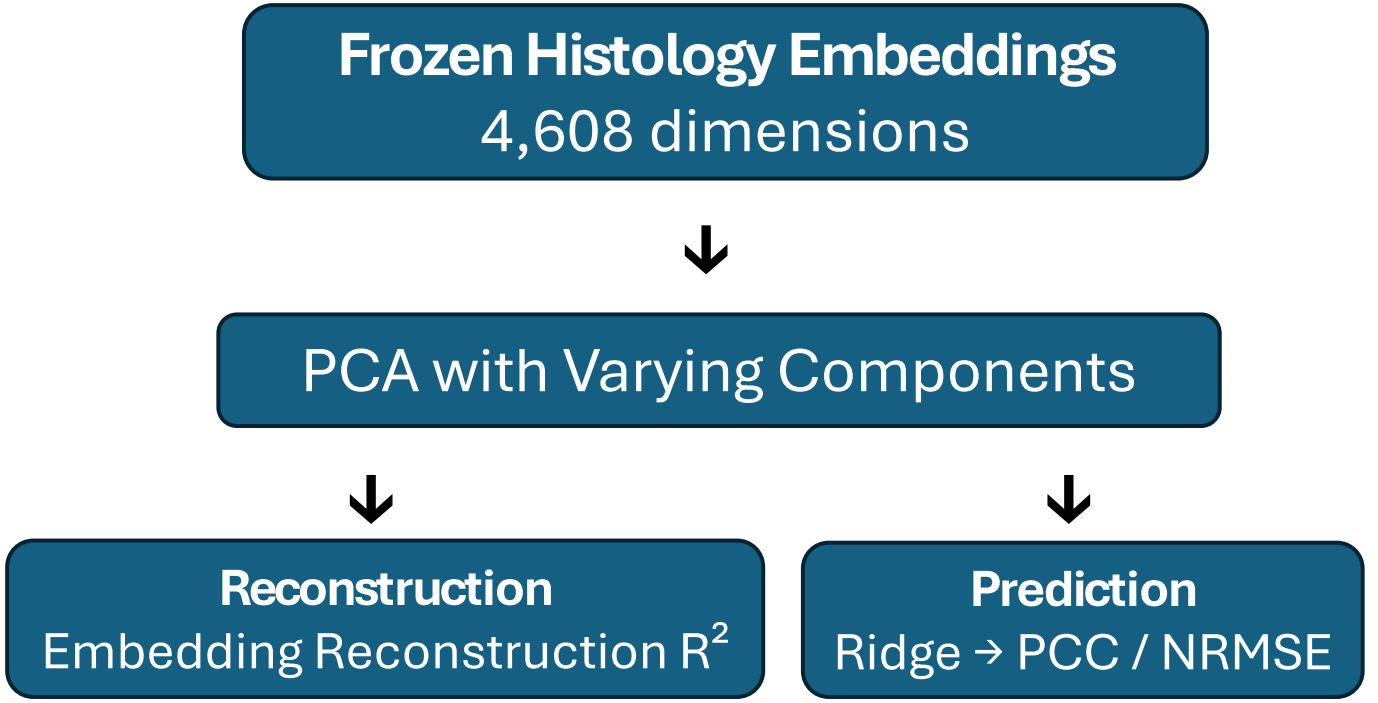
Current Status: Baseline and evaluation code successfully tested; comprehensive multi-patient evaluation remains pending.

PCA Compression: Reconstruction vs. Prediction

Ablation Study: Is PCA Necessary?

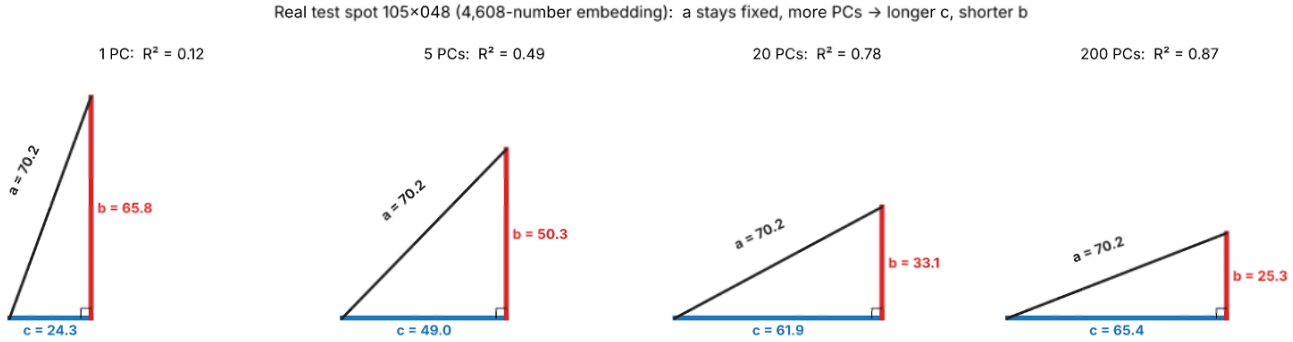
Preliminary Ablation Setup (TENX95)

- 7,592 matched H&E patches from one slide.
- PCA dimensions: 50, 100, 256, 500, 1000; plus No PCA.
- Initial random spot split followed by five-fold spatial-block validation.

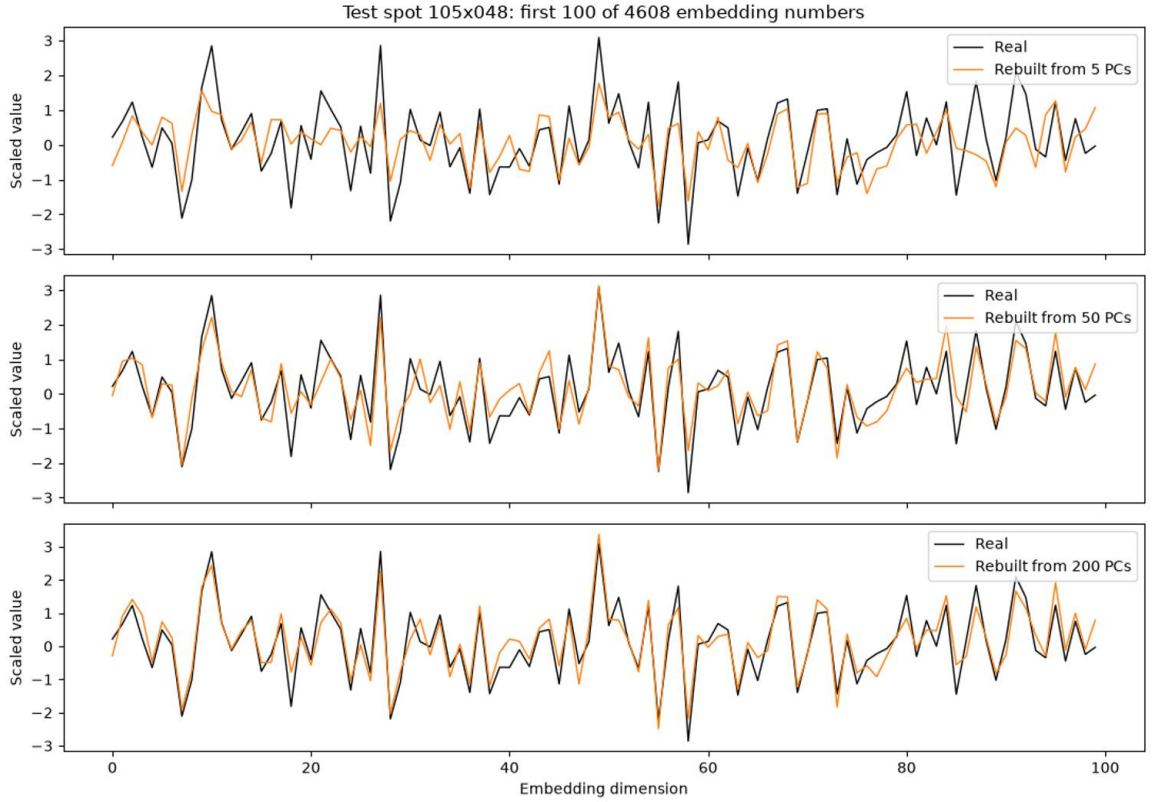


PCA Reconstruction

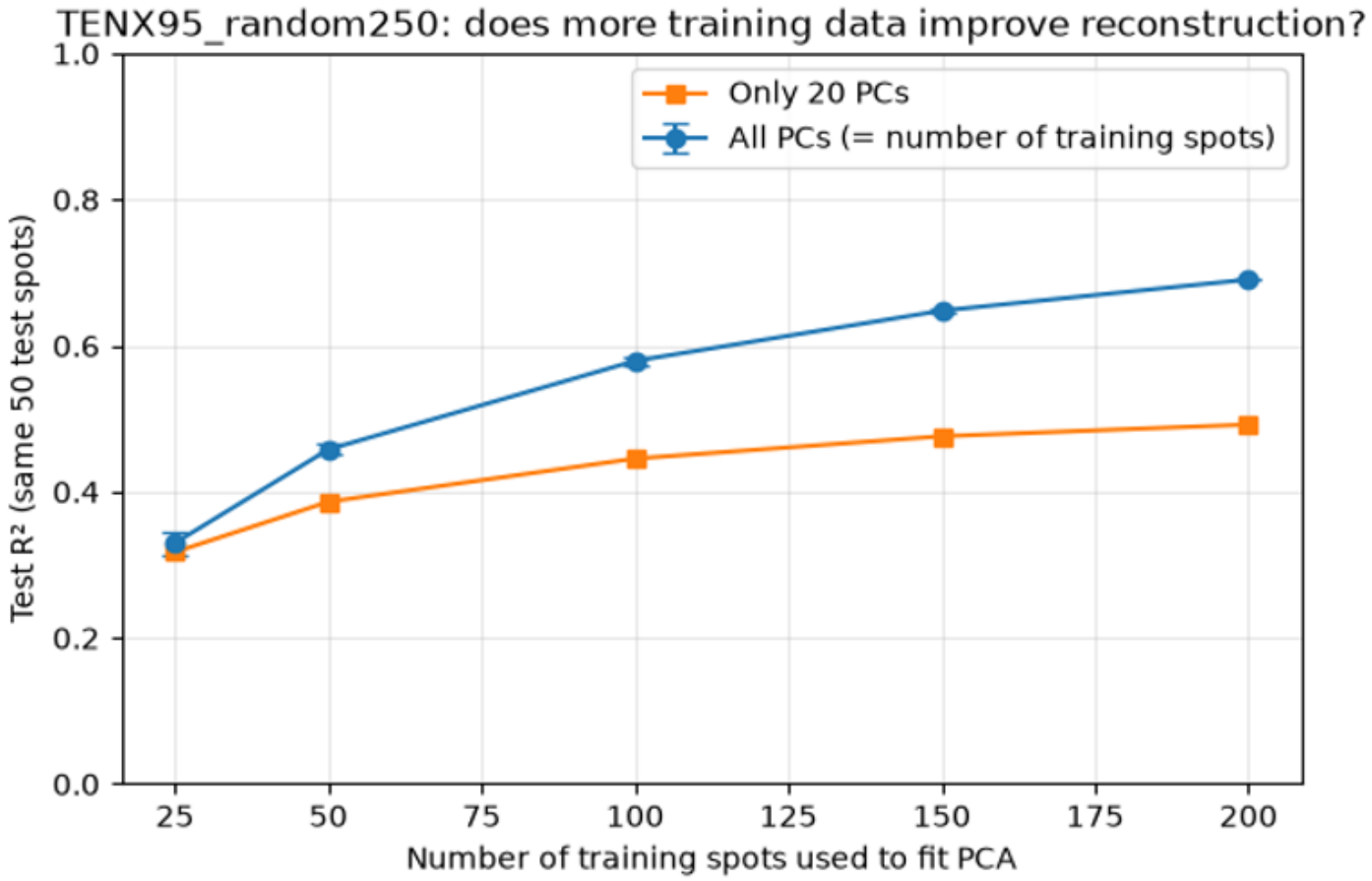
- For any k, rebuild each spot from its first k PCs.



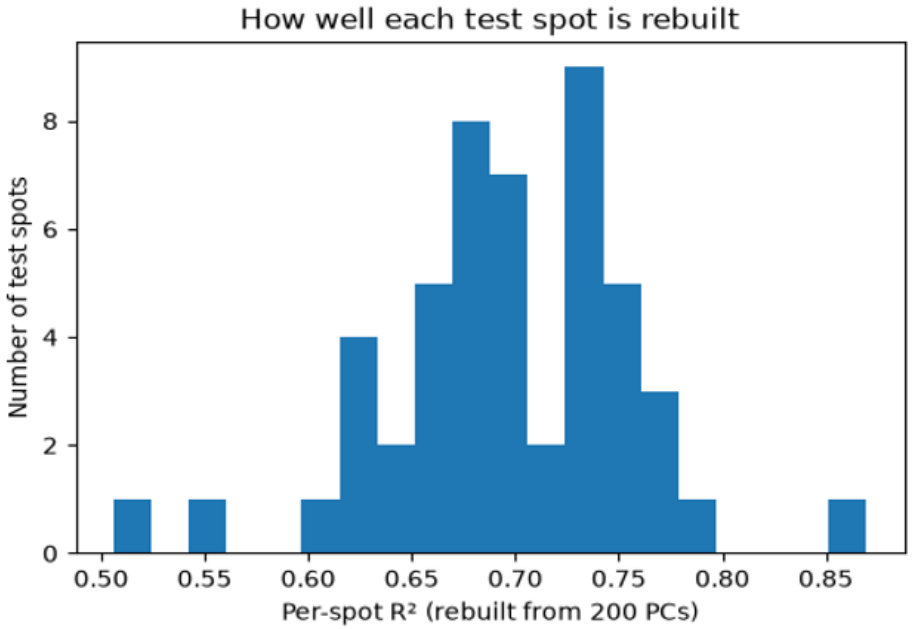
PCs (k)	Train R^2	Test R^2	Test cosine
1	0.137	0.130	0.323
2	0.224	0.208	0.426
5	0.358	0.305	0.529
10	0.477	0.392	0.615
20	0.610	0.492	0.691
50	0.784	0.582	0.758
100	0.910	0.642	0.800
150	0.970	0.671	0.820
200	1.000	0.691	0.833



PCA Reconstruction



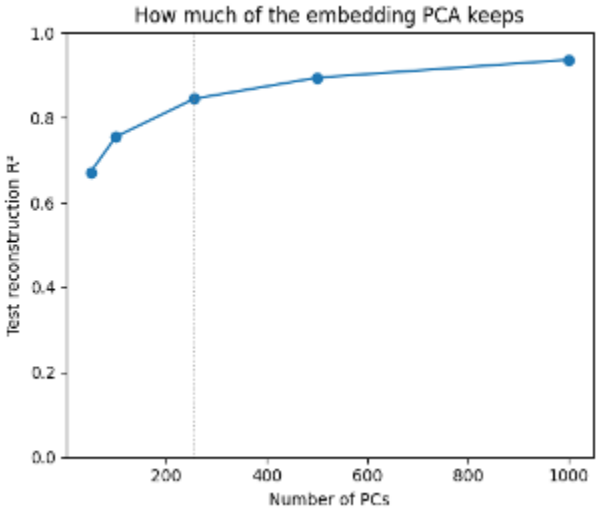
Training spots	Test R^2 (all PCs)	Test R^2 (20 PCs)
25	0.330 ± 0.015	0.317
50	0.458 ± 0.007	0.386
100	0.579 ± 0.005	0.445
150	0.648 ± 0.003	0.476
200	0.691	0.492

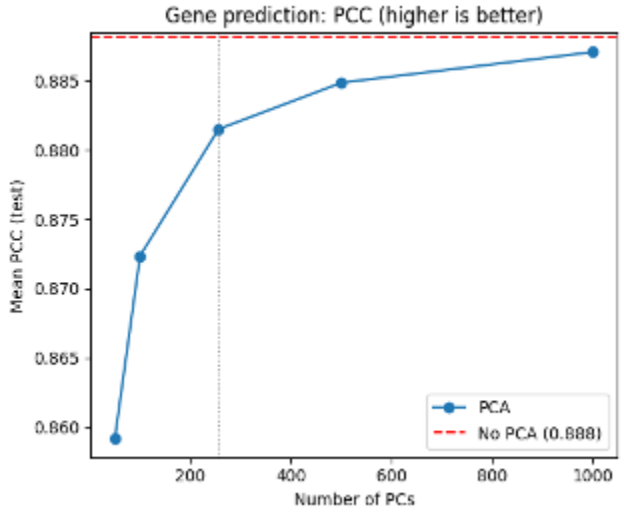


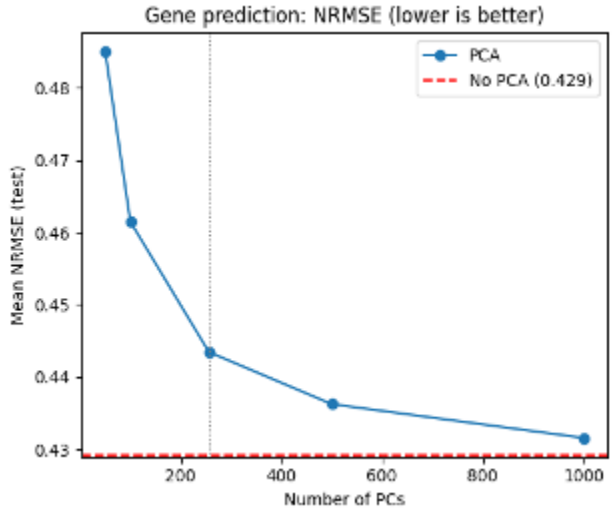
And continuing...

No PCA?

- Use **all 7,592 spots** on the slide. Each spot becomes 4,608 numbers.
- Split them at random: **6,073 train**, **1,519 test**.
- Pick the **50 genes** that vary most, using the train spots only.
- Make the inputs: PCA with 50, 100, 256, 500 or 1,000 PCs, or **No PCA** (all 4,608 numbers).
- For each input, train ridge on the train spots, then predict the 50 genes for the test spots.
- Score with **PCC**: how closely predicted genes match real genes (1 = perfect).



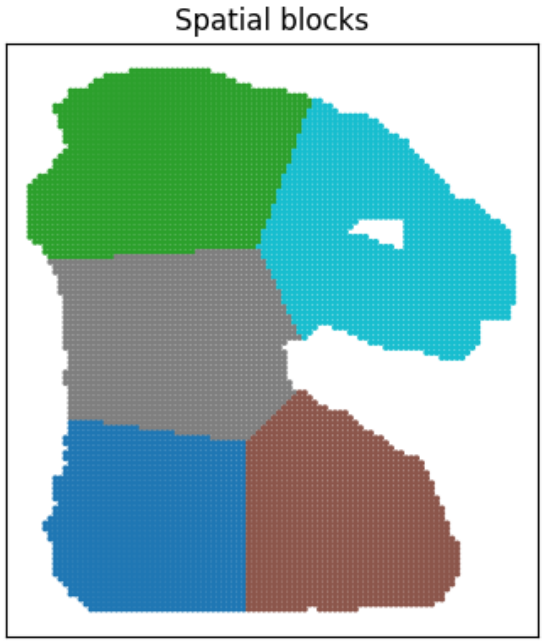


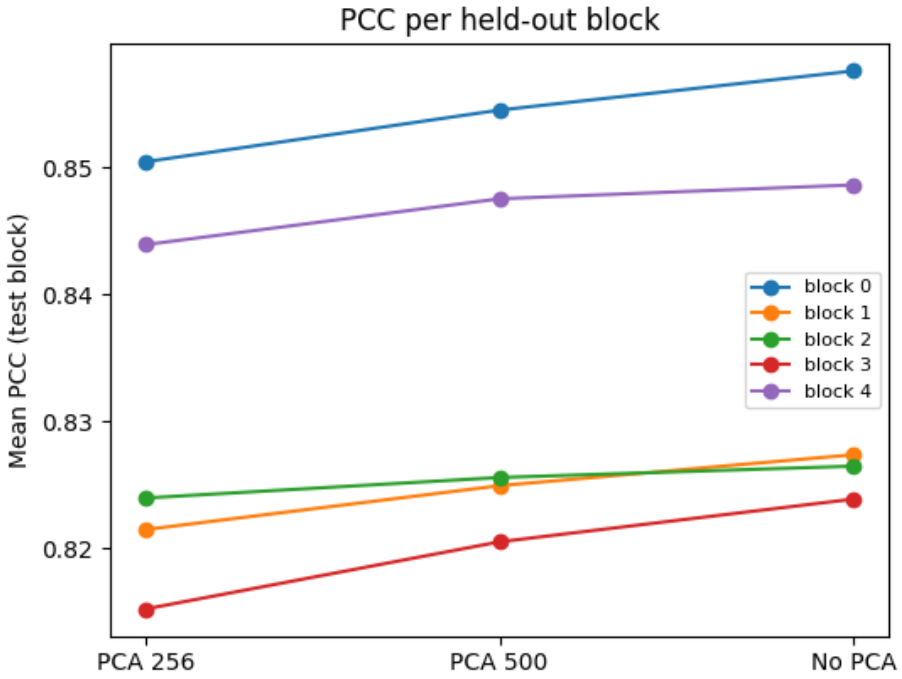


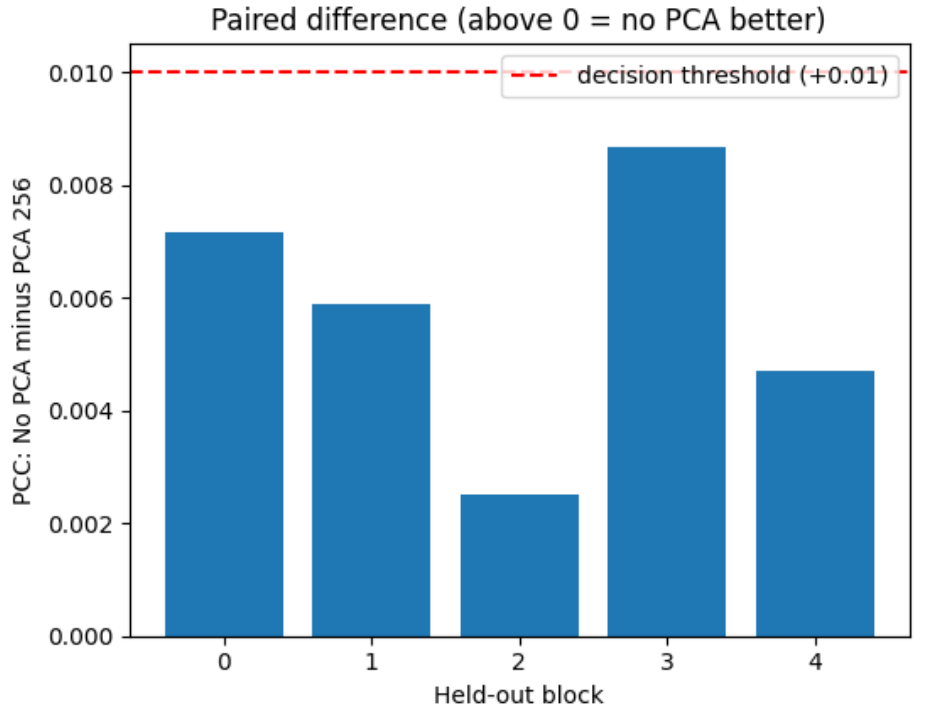
Setting	Recon R^2 (test)	PCC	Time
PCA 50	0.672	0.859	5 s
PCA 100	0.755	0.872	11 s
PCA 256	0.845	0.881	11 s
PCA 500	0.894	0.885	11 s
PCA 1000	0.936	0.887	16 s
No PCA	—	0.888	115 s

No PCA? Part 2. Trying Spatial Split

- Cut the slide into **5 regions**.
- Use **1 region as test** and the other 4 as train.
- Remove training spots **within 2 spots** of the test region.
- Pick ridge's λ using regions inside the training spots (not single spots, which also leaks).
- Repeat **5 times** so each region gets a turn as test. Compare PCA 256, PCA 500 and No PCA on the same spots each time.
- Rule set before running: drop PCA only if No PCA wins by **at least 0.01 PCC** and in **at least 4 of 5 regions**.

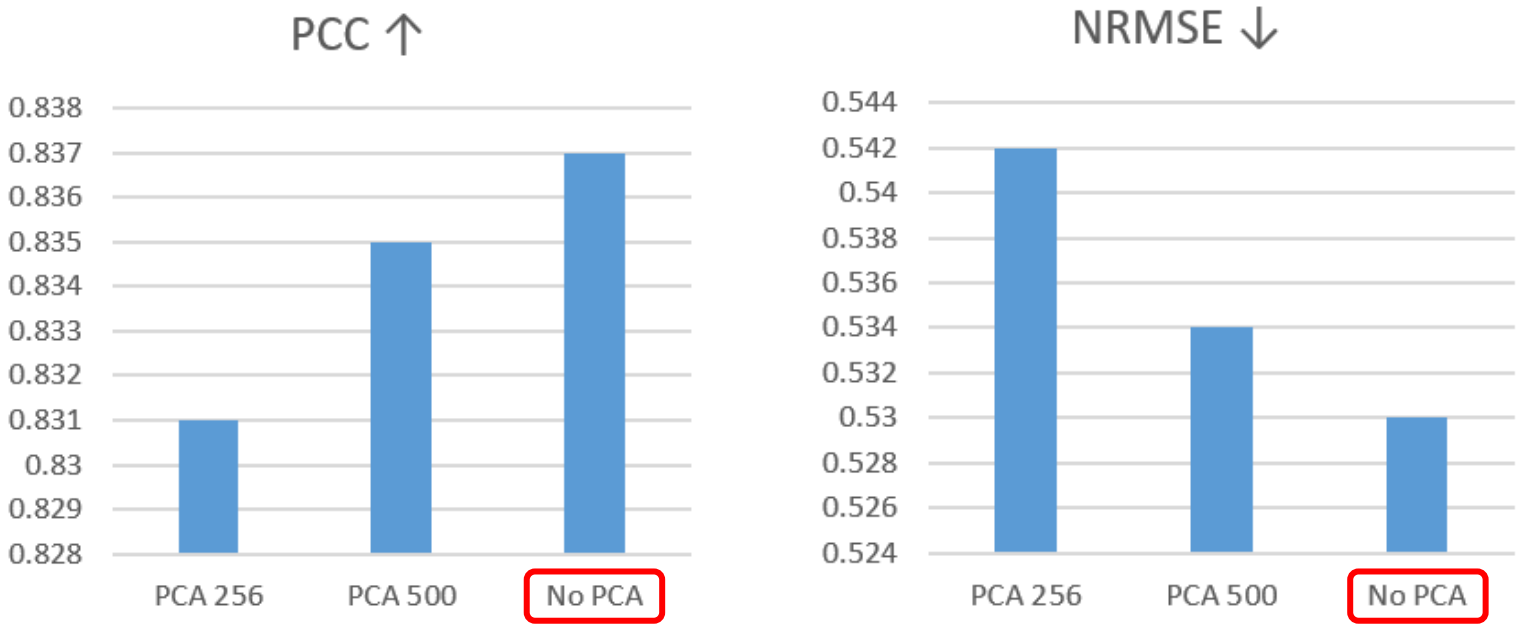






Preliminary Results: Accuracy vs. Efficiency

TENX95 — Five-Fold Spatial-Block Cross-Validation



Key Findings

- No PCA slightly improves prediction accuracy (+0.006 PCC).
- Full-feature Ridge is approximately 10× slower.
- PCA 500 offers a possible intermediate setting.

Preliminary Decision: Retain PCA 256 as the baseline, pending larger-scale validation.

Q & A

Group 5 | Project F

Spatial Gene Expression Prediction from Breast Histology

